

Erratum: “Bulk is Gauge” Was a Quantifier Error

The Joint Bulk Floor is Nonzero, and Larger Than the Ghost Mode

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Abstract

Observation 1 of the preceding note (“Bulk is gauge”) claimed that every tested bulk antidiagonal cell has minimal achievable spread exactly zero, and concluded that bulk routing carries no forced history-dependence. This conflated two different statements: that every cell *individually* admits a face point with zero spread, versus that *one* face point gives every cell zero spread simultaneously. The first is what was tested; the second is what the conclusion needed. Running the correct joint test – minimizing the worst-case spread across all bulk cells under a single shared point of the near-optimal face – gives a floor of 1.475854×10^{-2} at depth 3, an order of magnitude *larger* than the previously reported ghost-cell floor of 1.557×10^{-3} . Bulk routing is not gauge. Six of the seven tested cells sit at exactly the same spread value at the joint optimum, indicating a shared coupling constraint rather than seven independent phenomena.

1 The error

Observation 1 of the prior note stated: “At depth 4, for every tested (s, x) ... $\text{spread}^*(s, x) = 0$... Every bulk routing pattern previously observed to vary across histories was solver gauge, not forced structure.” The computation behind the first sentence is correct: each cell, tested in isolation, does admit a face point with zero spread. The second sentence does not follow from it. Symbolically, what was shown is

$$(\forall (s, x)) (\exists z_{(s, x)} \in F^*) \text{ spread}(s, x; z_{(s, x)}) = 0,$$

where F^* is the near-optimal face and $z_{(s, x)}$ may depend on (s, x) . What “bulk is gauge” requires is the quantifier-reversed statement

$$(\exists z \in F^*) (\forall (s, x)) \text{ spread}(s, x; z) = 0,$$

a single z working for every cell at once. These are not equivalent, and only the second licenses the conclusion that was drawn.

2 The corrected test

Proposition 1 (Joint face-spread LP). *For observables $q = 1, \dots, m$ (here, the tested bulk cells), the quantity*

$$\text{jointfloor}^* := \min_{z \in F^*} \max_q \left(\max_h L_{q, h}(z) - \min_h L_{q, h}(z) \right)$$

is computable by one linear program: introduce (U_q, L_q) per observable and a shared t with $U_q - L_q \leq t$ for every q , and minimize t subject to the same face constraint $|c^\top z - v^| \leq \tau$ used throughout. $\text{jointfloor}^* = 0$ correctly certifies that bulk is gauge; $\text{jointfloor}^* > 0$ certifies that it is not, regardless of what the per-cell minima were.*

3 Corrected result

Observation 1 (Retraction). *At depth 3, $M = 8$, testing the seven cells $s \in \{2, 3\}$ (all admissible x), every individual cell has $\text{spread}^*(s, x) = 0$, reproducing the earlier (correct, but insufficient) per-cell computation. The joint floor is*

$$\text{jointfloor}^* = 1.475854 \times 10^{-2},$$

*five orders of magnitude above the numerical tolerance. **Observation 1 of the prior note is retracted:** bulk routing is not gauge. No single point of the near-optimal face removes history-dependence from every bulk cell at once.*

Observation 2 (A shared bottleneck, not seven independent facts). *At the joint optimizer, six of the seven cells – $(s, x) \in \{(2, 0), (2, 1), (2, 2), (3, 1), (3, 2), (3, 3)\}$ – have spread exactly equal to the joint floor, 1.475854×10^{-2} ; the seventh, $(3, 0)$, has a strictly smaller spread, 4.983532×10^{-3} , at that same point. Six unrelated cells landing on the identical numerical value at the optimizer is not expected of seven independent obstructions; it is the signature of one shared coupling constraint (most plausibly a conservation or flow-consistency relation linking these cells’ routing) that the optimizer cannot relax without paying elsewhere. Identifying that constraint explicitly is more promising than treating the seven cells as separate data points.*

Remark 1 (Larger than the mode we were chasing). *The corrected bulk floor, 1.48×10^{-2} , is roughly ten times the ghost-cell floor, 1.557×10^{-3} , reported as the main finding of the prior note. The ghost cell was not wrong – it remains independently confirmed at two depths and two tolerances – but it is evidently not the dominant source of forced structure. The bulk coupling is.*

4 What remains to be done

The depth-4 joint bulk floor did not complete within the time budget available at the time of writing (the corresponding single-cell tests at depth 4 already ran to several hundred seconds each; the joint LP couples all cells into one larger program). Obtaining it, and comparing to the depth-3 value here, is the immediate next step: if the joint floor is roughly stable across depth (as the ghost cell’s floor was, Section 4 of the prior note), that is further evidence for a marginal, scale-stable mode; if it grows with depth, that is evidence for a relevant, dangerous one, in the renormalization language used elsewhere in this project.

Problem 16 (Identify and track the joint coupling). *Determine the linear relation among z responsible for pinning six of the seven tested cells to a common spread value, and track jointfloor^* across depth (at minimum depth 4, ideally depth 5) to determine whether this mode is bounded, shrinking, or growing with scale.*

5 Scope

This note corrects a specific, identified overreach in a previously circulated result and reports the outcome of the correct test in its place. It does not yet supply the depth-4 joint floor, does not identify the specific coupling constraint responsible for the six-way tie, and does not change the status of the ghost-cell finding, which stands independently of this correction.